McGill initiative in Computational Medicine

Intro To High-performance Computing

Workshop Lead: Megan Ng

Registration link:

Approximate duration: 2 hours

Prerequisites:

1. A Digital Research Alliance of Canada account
2. The Duo Mobile application downloaded on your phone
3. Globus Connect Personal software downloaded on your laptop
4. A code editor (The workshop lead will be using VS Code)

Summary:

This workshop is designed to give you practical experience using the high-performance computing system offered by the Digital Research Alliance of Canada (DRAC). Through guided hands-on activities, you can expect to leave this workshop with the knowledge and skills required run your own computational experiments using DRAC start to finish – from organizing and uploading your data and code to submitting jobs, troubleshooting common errors, and optimizing your workflow for efficient computing.

Learning Objectives:

1. Conceptually understand the workflow of using DRAC
2. Learn how to use DRAC start to finish
    a. Uploading data and code to DRAC storage systems
    b. Running code using DRAC high-performance computers
    c. Retrieving and assessing results
3. Identify and handle common errors that occur when using DRAC
4. Optimize local code for efficient use of DRAC

Content:

1. **Module I: About the Digital Research Alliance of Canada (10min)**
    a. What is the DRAC and the services they offer
    b. Why do researchers use DRAC services
    c. Workflow description of how you interact with DRAC's high-performance computing system

   d. Hands-on activity: Workflow understanding quiz (3min)

2. **Module II: Preparing your local data and code for high-performance computing (20min)**
   a. Data and code organization best practices
      i. Folder organization
      ii. Debugging locally
   b. How to communicate your computational needs to the high-performance computing system
      i. What is a job script
      ii. What information must you provide in a job script
   c. Hands-on activity: Editing job script template for personal use (10min)

3. **Module III: Uploading your data and code to DRAC's high-performance computing system (15min)**
   a. Introduction to DRAC's file transfer system (Globus)
   b. Alternate file transfer methods and their use cases
   c. Hands-on activity: Upload workshop data and code to DRAC using Globus Personal Connect (10min)
      i. Practice basic command line navigation commands (cd, ls, etc.) to confirm file transfer success

4. **Module IV: Running your code using DRAC's high-performance computing system (30min)**
   a. How to "submit a job"
   b. How to check on the status of your job
   c. How to interpret output files and error messages
   d. Hands-on activity: Submit a job, interpret output, address error (20min)

5. **Module V: Optimizing your code for high-performance computing (30min)**
   a. Overview of DRAC high-performance computing clusters
      i. How to choose the best cluster for your specific computational needs
   b. Submitting multiple jobs
      i. Introduction to bash scripting and parameter variables
   c. Hands-on activity: Edit your code to accommodate multi-job submissions (20min)